Chapter 7 Single-Dimensional Arrays

Opening Problem

Read one hundred numbers, compute their average, and find out how many numbers are above the average.

- Your program first reads the numbers and computes their average, then compares each number with the average to determine whether it is above the average.
- In order to accomplish this task, the numbers must all be stored in variables.
- You have to declare **100** variables and repeatedly write almost identical code 100 times.
- Writing a program this way would be impractical.

Opening Problem (cont.)

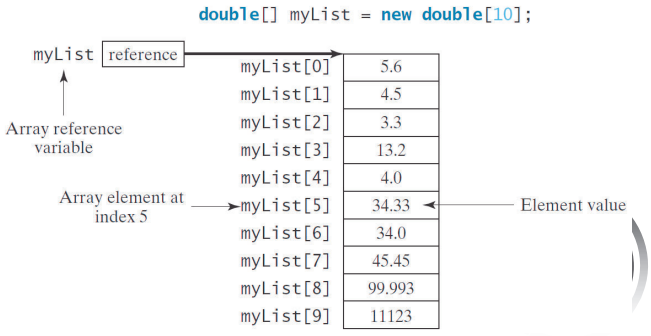
- How do you solve this problem?
- Java provides a data structure, the *array*, which stores a fixed-size sequential collection of elements of the same type.
- In this example, you can store all 100 numbers into an array and access them through a single array variable.

Objectives

- To declare array reference variables and create arrays
- To access array elements using indexes
- To declare, create, and initialize an array using an array initializer
- To program common array operations (displaying arrays, summing all elements, finding the minimum and maximum elements, random shuffling, and shifting elements)
- To simplify programming using the foreach loops
- To copy contents from one array to another
- To develop and invoke methods with array arguments and return values
- To define a method with a variable-length argument list
- To search elements using the linear or binary search algorithm.
- To sort an array using the selection sort approach
- To use the methods in the **java.util.Arrays** class

Introducing Arrays

Array is a data structure that represents a collection of the same types of data.



Declaring Array Variables

- You must declare a variable to reference the array and specify the array's *element type*.
  Here is the syntax for declaring an array variable:
- `datatype[] arrayRefVar;`

  Example:

  `double[] myList;`
- `datatype arrayRefVar[];` // This style is allowed, but not preferred

  Example:

  `double myList[];`

Creating Arrays

- After an array variable is declared, you can create an array by using the **new** operator and assign its reference to the variable with the following syntax:

```
arrayRefVar = new datatype[arraySize];
```

Example:

```
myList = new double[10];
```

`myList[0]` references the first element in the array.

`myList[9]` references the last element in the array.

Declaring and Creating in One Step

- `datatype[] arrayRefVar = new datatype[arraySize];`

```
double[] myList = new double[10];
```

- `datatype arrayRefVar[] = new datatype[arraySize];`

```
double myList[] = new double[10];
```

The Length of an Array

Once an array is created, its size is fixed. It cannot be changed. You can find its size using

```
arrayRefVar.length
```

For example,

`myList.length` returns 10



Default Values

When an array is created, its elements are assigned the default value of

0 for the numeric primitive data types,
'\u0000' for char types, and
false for boolean types.



Indexed Variables

The array elements are accessed through the index. The array indices are *0-based*, i.e., it starts from 0 to arrayRefVar.length-1.

In the example, myList holds ten double values and the indices are from 0 to 9.

Each element in the array is represented using the following syntax, known as an *indexed variable*:

```
arrayRefVar[index];
```



Using Indexed Variables

To assign values to the elements, use the syntax:

```
arrayRefVar[index] = value;
```

Example:

```
myList[5] = 20.5;
```



Using Indexed Variables (cont.)

After an array is created, an indexed variable can be used in the same way as a regular variable.

For example, the following code adds the value in myList[0] and myList[1] to myList[2].

```
myList[2] = myList[0] + myList[1];
```

Example:

```
for (int i = 0; i < myList.length; i++)
{
    myList[i] = i;
}
```



Array Initializers

Declaring, creating, initializing in one step:

```
elementType[] arrayRefVar = {value0,
value1, ..., valuek};
```

Example:

```
double[] myList = {1.9, 2.9, 3.4, 3.5};
```

This shorthand syntax must be in one statement.



Declaring, creating, initializing Using the Shorthand Notation

```
double[] myList = {1.9, 2.9, 3.4, 3.5};
```

This shorthand notation is equivalent to the following statements:

```
double[] myList = new double[4];
myList[0] = 1.9;
myList[1] = 2.9;
myList[2] = 3.4;
myList[3] = 3.5;
```



CAUTION

- Using the shorthand notation, you have to declare, create, and initialize the array all in one statement.
- Splitting it would cause a syntax error.
- For example, the following is wrong:

```
double[] myList;

myList = {1.9, 2.9, 3.4, 3.5};
```



animation

Trace Program with Arrays

Declare array variable values, create an array, and assign its reference to values

```
public class Test {
  public static void main(String[] args) {
    int[] values = new int[5];
    for (int i = 1; i < 5; i++) {
      values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[4];
  }
}
```

After the array is created

0	0
1	0
2	0
3	0
4	0



animation

Trace Program with Arrays

i becomes 1

```
public class Test {
  public static void main(String[] args) {
    int[] values = new int[5];
    for (int i = 1; i < 5; i++) {
      values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[4];
  }
}
```

After the array is created

0	0
1	0
2	0
3	0
4	0



animation

Trace Program with Arrays

i (=1) is less than 5

```
public class Test {
  public static void main(String[] args) {
    int[] values = new int[5];
    for (int i = 1; i < 5; i++) {
      values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[4];
  }
}
```

After the array is created

0	0
1	0
2	0
3	0
4	0



animation

Trace Program with Arrays

After this line is executed, value[1] is 1

```
public class Test {
  public static void main(String[] args) {
    int[] values = new int[5];
    for (int i = 1; i < 5; i++) {
      values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[4];
  }
}
```

After the first iteration

0	0
1	1
2	0
3	0
4	0



animation

Trace Program with Arrays

After i++, i becomes 2

```
public class Test {
  public static void main(String[] args) {
    int[] values = new int[5];
    for (int i = 1; i < 5; i++) {
      values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[4];
  }
}
```

After the first iteration

0	0
1	1
2	0
3	0
4	0



animation

Trace Program with Arrays

i (= 2) is less than 5

```
public class Test {
  public static void main(String[]
      args) {
    int[] values = new int[5];
    for (int i = 1; i < 5; i++) {
      values[i] = i + values[i-1];
    }
    values[0] = values[1] +
      values[4];
  }
}
```

After the first iteration

0	0
1	1
2	0
3	0
4	0



animation

Trace Program with Arrays

After this line is executed, values[2] is 3 (2 + 1)

```
public class Test {
  public static void main(String[] args) {
    int[] values = new int[5];
    for (int i = 1; i < 5; i++) {
      values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[4];
  }
}
```

After the second iteration

0	0
1	1
2	3
3	0
4	0



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Trace Program with Arrays

After this, i becomes 3.

```
public class Test {
  public static void main(String[] args) {
    int[] values = new int[5];
    for (int i = 1; i < 5; i++) {
      values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[4];
  }
}
```

After the second iteration

0	0
1	1
2	3
3	0
4	0



animation

Trace Program with Arrays

i (=3) is still less than 5.

```
public class Test {
  public static void main(String[] args) {
    int[] values = new int[5];
    for (int i = 1; i < 5; i++) {
      values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[4];
  }
}
```

After the second iteration

0	0
1	1
2	3
3	0
4	0



animation

Trace Program with Arrays

After this line, values[3] becomes 6 (3 + 3)

```
public class Test {
  public static void main(String[] args) {
    int[] values = new int[5];
    for (int i = 1; i < 5; i++) {
      values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[4];
  }
}
```

After the third iteration

0	0
1	1
2	3
3	6
4	0



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Trace Program with Arrays

After this, i becomes 4

```
public class Test {
  public static void main(String[] args) {
    int[] values = new int[5];
    for (int i = 1; i < 5; i++) {
      values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[4];
  }
}
```

After the third iteration

0	0
1	1
2	3
3	6
4	0



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Trace Program with Arrays

i (=4) is still less than 5

```
public class Test {
  public static void main(String[] args) {
    int[] values = new int[5];
    for (int i = 1; i < 5; i++) {
      values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[4];
  }
}
```

After the third iteration

0	0
1	1
2	3
3	6
4	0



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Trace Program with Arrays

After this, values[4] becomes 10 (4 + 6)

```
public class Test {
  public static void main(String[] args) {
    int[] values = new int[5];
    for (int i = 1; i < 5; i++) {
      values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[4];
  }
}
```

After the fourth iteration

0	0
1	1
2	3
3	6
4	10



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Trace Program with Arrays

After i++, i becomes 5

```
public class Test {
  public static void main(String[] args) {
    int[] values = new int[5];
    for (int i = 1; i < 5; i++) {
      values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[4];
  }
}
```

After the fourth iteration

0	0
1	1
2	3
3	6
4	10



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Trace Program with Arrays

i (=5) < 5 is false. Exit the loop

```
public class Test {
  public static void main(String[] args) {
    int[] values = new int[5];
    for (int i = 1; i < 5; i++) {
      values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[4];
  }
}
```

After the fourth iteration

0	0
1	1
2	3
3	6
4	10



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Trace Program with Arrays

After this line, values[0] is 11 (1 + 10)

```
public class Test {
  public static void main(String[] args) {
    int[] values = new int[5];
    for (int i = 1; i < 5; i++) {
      values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[4];
  }
}
```

0	11
1	1
2	3
3	6
4	10



Errors

```
public class Test {
  public static void main(String[] args)
  {
    int[] values = new int[5];
    for (int i = 1; i < 6; i++) {
       values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[4];
  }
}
```

Runtime Error: ArrayIndexOutOfBoundsException: 5



Errors (cont.)

```
public class Test {
  public static void main(String[] args)
  {
    int[] values = new int[5];
    for (int i = 1; i < 6; i++) {
       values[i] = i + values[i-1];
    }
    values[0] = values[1] + values[5];
  }
}
```

Runtime Error: ArrayIndexOutOfBoundsException: 5



Processing Arrays

See the examples in the text.

1. (Initializing arrays with input values)
2. (Initializing arrays with random values)
3. (Printing arrays)
4. (Summing all elements)
5. (Finding the largest element)
6. (Finding the smallest index of the largest element)
7. (*Random shuffling*)
8. (*Shifting elements*)



Initializing arrays with input values

```
double[] myList = new double[5];

Scanner input = new Scanner(System.in);
System.out.print("Enter " + myList.length + "
   values: ");

for (int i = 0; i < myList.length; i++)
  myList[i] = input.nextDouble();
```



Initializing arrays with random values

```
for (int i = 0; i < myList.length; i++) {
  myList[i] = Math.random() * 100;
}
```



Printing arrays

```
for (int i = 0; i < myList.length; i++){
  System.out.print(myList[i] + " ");
}
```



Summing all elements

```
double total = 0;
for (int i = 0; i < myList.length; i++){
  total += myList[i];
}
```



Finding the largest element

```
double max = myList[0];
for (int i = 1; i < myList.length; i++){
  if (myList[i] > max)
    max = myList[i];
}
```



Finding the smallest index of the largest element

// Suppose the array **myList** is {**1**, **5**, **3**, **4**, **5**, **5**}.

```
double max = myList[0];
int indexOfMax = 0;

for (int i = 1; i < myList.length; i++){
    if (myList[i] > max) {
      max = myList[i];
      indexOfMax = i;
    }
}
```

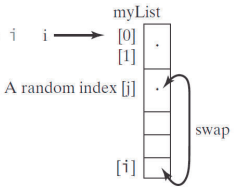


Random shuffling

In many applications, you need to randomly reorder the elements in an array. This is called *shuffling*.

```
for (int i = myList.length - 1; i > 0; i--) {
  // Generate an index j randomly with 0 <= j <= i
  int j = (int)(Math.random()
    * (i + 1));

  // Swap myList[i] with myList[j]
  double temp = myList[i];
  myList[i] = myList[j];
  myList[j] = temp;
}
```





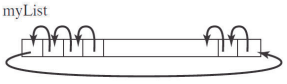
Shifting Elements

Sometimes you need to shift the elements left or right.

```
double temp = myList[0]; // Retain the first element

// Shift elements left
for (int i = 1; i < myList.length; i++) {
  myList[i - 1] = myList[i];
}

// Move the first element to fill in the last position
myList[myList.length - 1] = temp;
```





Simplifying coding

```
if (monthNumber == 1)
    System.out.println("The month is January");

else if (monthNumber == 2)
    System.out.println("The month is February");
...
else
    System.out.println("The month is December");
```



Simplifying coding

```
String[] months = {"January", "February", ..., "December"};

System.out.print("Enter a month number (1 to 12): ");

int monthNumber = input.nextInt();

System.out.println("The month is " + months[monthNumber - 1]);
```



Enhanced for Loop (for-each loop)

Java supports a convenient **for** loop, known as a *foreach loop*, which enables you to traverse the array sequentially without using an index variable.

For example, the following code displays all elements in the array myList:

```
for (double value: myList)
    System.out.println(value);
```



Enhanced for Loop (for-each loop)

In general, the syntax is

```
for (elementType value: arrayRefVar) {
    // Process the value
}
```

You still have to use an index variable if you wish to traverse the array in a different order or change the elements in the array.



Example 1

```
int x = 30;
int[] numbers = new int[x];
x = 60;
System.out.println("x is " + x);
System.out.println("The size of numbers
is " + numbers.length);
```

```
Output:
x is 60
The size of numbers is 30
```



Opening Problem

Read one hundred numbers, compute their average, and find out how many numbers are above the average.



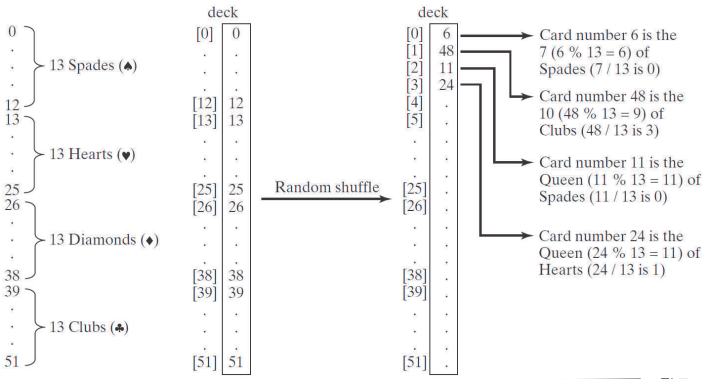
Problem: Deck of Cards

The problem is to write a program that picks four cards randomly from a deck of 52 cards. All the cards can be represented using an array named deck, filled with initial values 0 to 51, as follows:

```
int[] deck = new int[52];
// Initialize cards
for (int i = 0; i < deck.length; i++)
  deck[i] = i;
```



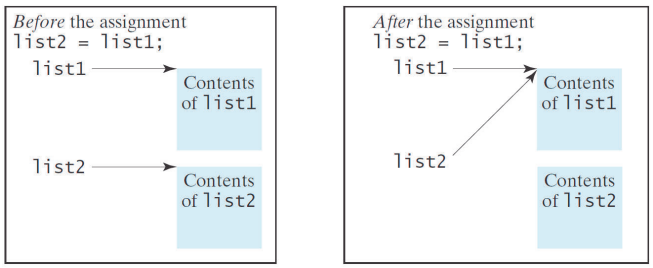
Problem: Deck of Cards, cont.



Copying Arrays

This statement does not copy the contents of the array referenced by **list1** to **list2**, but instead merely copies the reference value from **list1** to **list2**. After this statement, **list1** and **list2** reference the same array,

list2 = list1;



Copying Arrays

There are three ways to copy arrays:

- Use a loop to copy individual elements one by one.
- Use the static **arraycopy** method in the **System** class.
- Use the **clone** method to copy arrays; this will be introduced in Chapter 13, Abstract Classes and Interfaces.

Copying Arrays

Using a loop:

```
int[] sourceArray = {2, 3, 1, 5, 10};
int[] targetArray = new
  int[sourceArray.length];

for (int i = 0; i < sourceArrays.length; i++)
    targetArray[i] = sourceArray[i];
```


The `arraycopy` Utility

```
arraycopy(sourceArray, src_pos,
  targetArray, tar_pos, length);
```

- The parameters **srcPos** and **tarPos** indicate the starting positions in **sourceArray** and **targetArray**, respectively. The number of elements copied from **sourceArray** to **targetArray** is indicated by **length**.

Example:

```
System.arraycopy(sourceArray, 0,
  targetArray, 0, sourceArray.length);
```


Passing Arrays to Methods

```
public static void printArray(int[] array) {
  for (int i = 0; i < array.length; i++) {
    System.out.print(array[i] + " ");
  }
}
```

Invoke the method

```
int[] list = {3, 1, 2, 6, 4, 2};
printArray(list);
```

Invoke the method

```
printArray(new int[]{3, 1, 2, 6, 4, 2});
```

Anonymous array

Anonymous Array

The statement

```
printArray(new int[]{3, 1, 2, 6, 4, 2});
```

creates an array using the following syntax:

```
new dataType[]{literal0, literal1, ..., literalk};
```

There is no explicit reference variable for the array. Such array is called an *anonymous array*.

Pass By Value

Java uses *pass by value* to pass arguments to a method. There are important differences between passing a value of variables of primitive data types and passing arrays.

- **For a parameter of a primitive type value**, the actual value is passed. Changing the value of the local parameter inside the method does not affect the value of the variable outside the method.
- **For a parameter of an array type**, the value of the parameter contains a reference to an array; this reference is passed to the method. Any changes to the array that occur inside the method body will affect the original array that was passed as the argument.

Simple Example

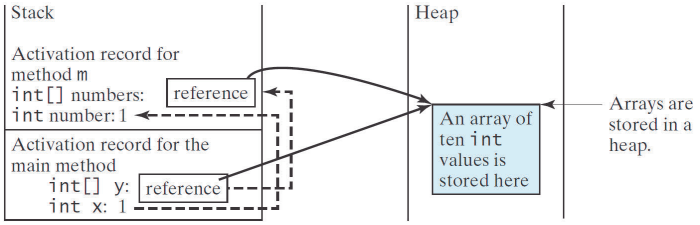
```
public class Test {
  public static void main(String[] args) {
    int x = 1; // x represents an int value
    int[] y = new int[10]; // y represents an array of int values

    m(x, y); // Invoke m with arguments x and y

    System.out.println("x is " + x);
    System.out.println("y[0] is " + y[0]);
  }

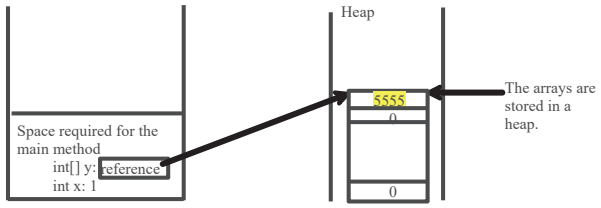
  public static void m(int number, int[] numbers) {
    number = 1001; // Assign a new value to number
    numbers[0] = 5555; // Assign a new value to numbers[0]
  }
}
```


Call Stack



When invoking m(x, y), the values of x and y are passed to number and numbers. Since y contains the reference value to the array, numbers now contains the same reference value to the same array.

Heap



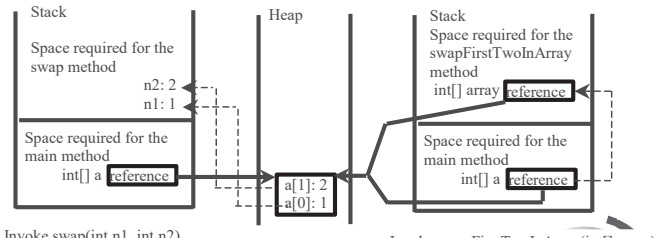
The JVM stores the array in an area of memory, called *heap*, which is used for dynamic memory allocation where blocks of memory are allocated and freed in an arbitrary order.

Passing Arrays as Arguments

- Objective: Demonstrate differences of passing primitive data type variables and array variables.

TestPassArray Run

Example, cont.



Invoke swap(int n1, int n2). The primitive type values in a[0] and a[1] are passed to the swap method.

The arrays are stored in a heap.

Invoke swapFirstTwoInArray(int[] array). The reference value in a is passed to the swapFirstTwoInArray method.

Returning an Array from a Method

```
public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

list

result

```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);
```

When a method returns an array, the reference of the array is returned

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Trace the reverse Method

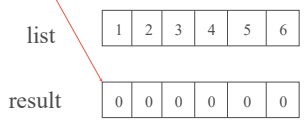
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

Declare result and create array





animation

Trace the reverse Method, cont.

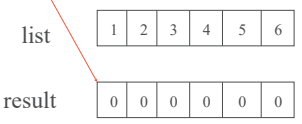
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

i = 0 and j = 5





animation

Trace the reverse Method, cont.

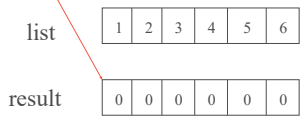
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

i (= 0) is less than 6





animation

Trace the reverse Method, cont.

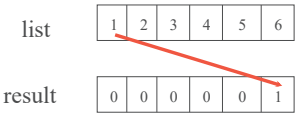
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

i = 0 and j = 5
Assign list[0] to result[5]





animation

Trace the reverse Method, cont.

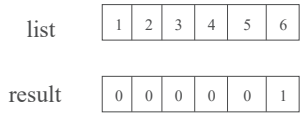
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

After this, i becomes 1 and j becomes 4





animation

Trace the reverse Method, cont.

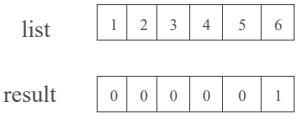
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

i (=1) is less than 6





animation

Trace the reverse Method, cont.

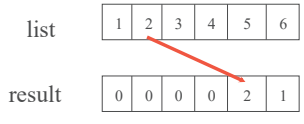
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

i = 1 and j = 4
Assign list[1] to result[4]





animation

Trace the reverse Method, cont.

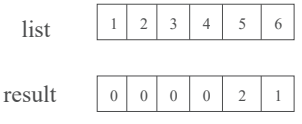
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

After this, i becomes 2 and j becomes 3





animation

Trace the reverse Method, cont.

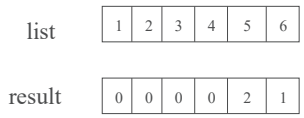
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

i (=2) is still less than 6





animation

Trace the reverse Method, cont.

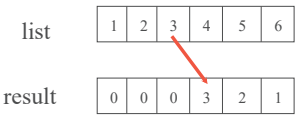
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

i = 2 and j = 3
Assign list[i] to result[j]





animation

Trace the reverse Method, cont.

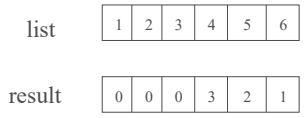
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

After this, i becomes 3 and
j becomes 2





animation

Trace the reverse Method, cont.

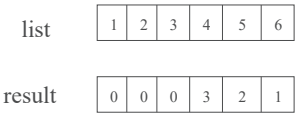
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

i (=3) is still less than 6





animation

Trace the reverse Method, cont.

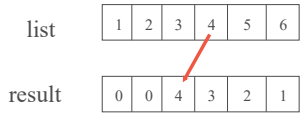
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

i = 3 and j = 2
Assign list[i] to result[j]





animation

Trace the reverse Method, cont.

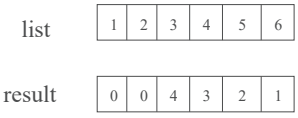
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

After this, i becomes 4 and
j becomes 1





animation

Trace the reverse Method, cont.

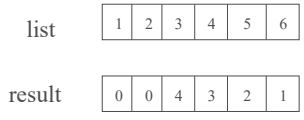
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

i (=4) is still less than 6





animation

Trace the reverse Method, cont.

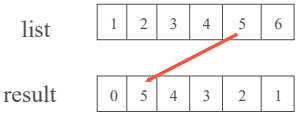
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

public static int[] reverse(int[] list) {
  int[] result = new int[list.length];

  for (int i = 0, j = result.length - 1;
       i < list.length; i++, j--) {
    result[j] = list[i];
  }

  return result;
}
```

i = 4 and j = 1
Assign list[i] to result[j]





animation

Trace the reverse Method, cont.

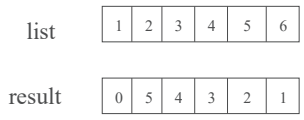
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

  public static int[] reverse(int[] list) {
    int[] result = new int[list.length];

    for (int i = 0, j = result.length - 1;
         i < list.length; i++, j--) {
      result[j] = list[i];
    }

    return result;
  }
```

After this, i becomes 5 and j becomes 0



animation

Trace the reverse Method, cont.

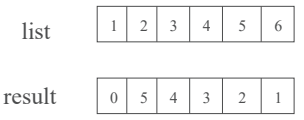
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

  public static int[] reverse(int[] list) {
    int[] result = new int[list.length];

    for (int i = 0, j = result.length - 1;
         i < list.length; i++, j--) {
      result[j] = list[i];
    }

    return result;
  }
```

i (=5) is still less than 6



animation

Trace the reverse Method, cont.

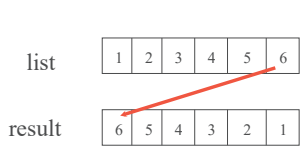
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

  public static int[] reverse(int[] list) {
    int[] result = new int[list.length];

    for (int i = 0, j = result.length - 1;
         i < list.length; i++, j--) {
      result[j] = list[i];
    }

    return result;
  }
```

i = 5 and j = 0
Assign list[i] to result[j]



animation

Trace the reverse Method, cont.

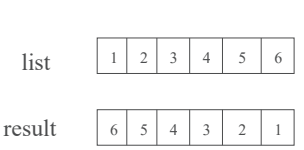
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

  public static int[] reverse(int[] list) {
    int[] result = new int[list.length];

    for (int i = 0, j = result.length - 1;
         i < list.length; i++, j--) {
      result[j] = list[i];
    }

    return result;
  }
```

After this, i becomes 6 and j becomes -1



animation

Trace the reverse Method, cont.

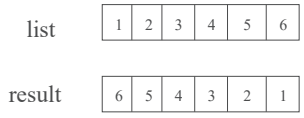
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

  public static int[] reverse(int[] list) {
    int[] result = new int[list.length];

    for (int i = 0, j = result.length - 1;
         i < list.length; i++, j--) {
      result[j] = list[i];
    }

    return result;
  }
```

i (=6) < 6 is false. So exit the loop.



animation

Trace the reverse Method, cont.

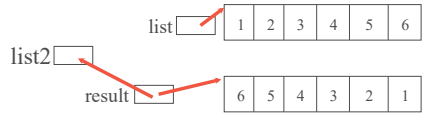
```
int[] list1 = {1, 2, 3, 4, 5, 6};
int[] list2 = reverse(list1);

  public static int[] reverse(int[] list) {
    int[] result = new int[list.length];

    for (int i = 0, j = result.length - 1;
         i < list.length; i++, j--) {
      result[j] = list[i];
    }

    return result;
  }
```

Return result



Problem: Counting Occurrence of Each Letter

- Generate 100 lowercase letters randomly and assign to an array of characters.
- Count the occurrence of each letter in the array.

CountLettersInArray Run

Searching Arrays

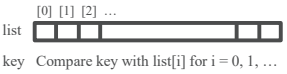
- Searching is the process of looking for a specific element in an array; for example, discovering whether a certain score is included in a list of scores.
- Searching is a common task in computer programming.
- There are many algorithms and data structures devoted to searching.
- In this section, two commonly used approaches are discussed,
  - *Linear search*
  - *Binary search*

Linear Search

- The linear search approach compares the key element, key, *sequentially* with each element in the array list.
- The method continues to do so until the key matches an element in the list or the list is exhausted without a match being found.
- If a match is made, the linear search returns the index of the element in the array that matches the key.
- If no match is found, the search returns -1.

Linear Search

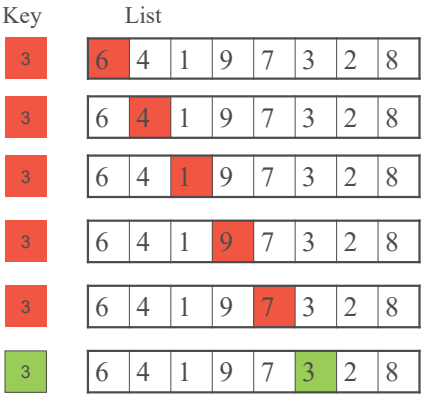
```
public class LinearSearch {
  /** The method for finding a key in the list */
  public static int linearSearch(int[] list, int key) {
    for (int i = 0; i < list.length; i++)
      if (key == list[i])
        return i;
    return -1;
  }
}
```



key Compare key with list[i] for i = 0, 1, …

animation

Linear Search Animation



From Idea to Solution

```
/** The method for finding a key in the list */
public static int linearSearch(int[] list, int key) {
  for (int i = 0; i < list.length; i++)
    if (key == list[i])
      return i;
  return -1;
}
```

Trace the method

```
int[] list = {1, 4, 4, 2, 5, -3, 6, 2};
int i = linearSearch(list, 4);  // returns 1
int j = linearSearch(list, -4); // returns -1
int k = linearSearch(list, -3); // returns 5
```


Binary Search

For binary search to work, the elements in the array must already be ordered. Without loss of generality, assume that the array is in ascending order.

e.g., 2 4 7 10 11 45 50 59 60 66 69 70 79

The binary search first compares the key with the element in the middle of the array.

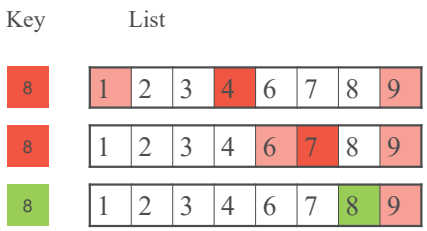
Binary Search, cont.

Consider the following three cases:

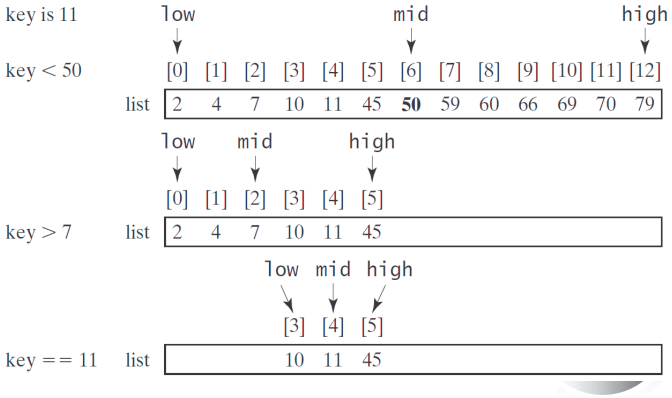
- If the key is less than the middle element, you only need to search the key in the first half of the array.
- If the key is equal to the middle element, the search ends with a match.
- If the key is greater than the middle element, you only need to search the key in the second half of the array.

animation

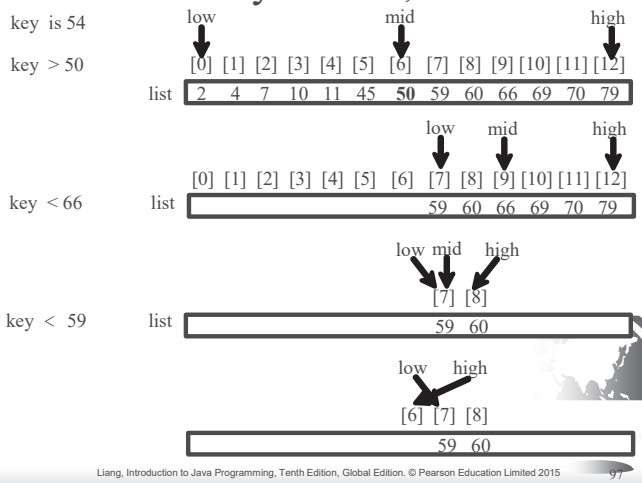
Binary Search



Binary Search, cont.



Binary Search, cont.



Binary Search, cont.

The binarySearch method returns the index of the element in the list that matches the search key if it is contained in the list. Otherwise, it returns

-insertion point - 1.

The insertion point is the point at which the key would be inserted into the list.

From Idea to Soluton

```
/** Use binary search to find the key in the list */
public static int binarySearch(int[] list, int key) {
  int low = 0;
  int high = list.length - 1;

  while (high >= low) {
    int mid = (low + high) / 2;
    if (key < list[mid])
      high = mid - 1;
    else if (key == list[mid])
      return mid;
    else
      low = mid + 1;
  }

  return -1 - low; // Now high < low, key not found
}
```

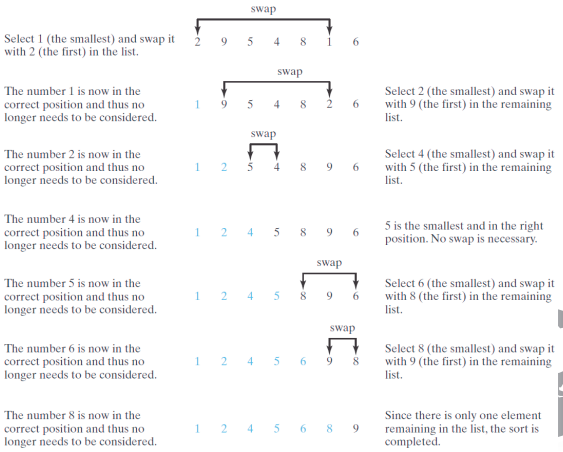

Sorting Arrays

- Sorting, like searching, is also a common task in computer programming.
- Many different algorithms have been developed for sorting.
- This section introduces a simple, intuitive sorting algorithms: *selection sort*.

Selection Sort

- Selection sort finds the smallest number in the list and places it first.
- It then finds the smallest number remaining and places it second, and so on until the list contains only a single number.

Selection Sort



From Idea to Solution

```
for (int i = 0; i < list.length; i++) {
  select the smallest element in list[i..listSize-1];
  swap the smallest with list[i], if necessary;
  // list[i] is in its correct position.
  // The next iteration apply on list[i..listSize-1]
}
```

```
list[0] list[1] list[2] list[3] ...                list[10]

list[0] list[1] list[2] list[3] ...                list[10]

list[0] list[1] list[2] list[3] ...                list[10]

list[0] list[1] list[2] list[3] ...                list[10]

list[0] list[1] list[2] list[3] ...                list[10]

                                  ...

list[0] list[1] list[2] list[3] ...                list[10]
```



```
for (int i = 0; i < listSize; i++) {
  select the smallest element in list[i..listSize-1];
  swap the smallest with list[i], if necessary;
  // list[i] is in its correct position.
  // The next iteration apply on list[i..listSize-1]
}
```

Expand

```
double currentMin = list[i];
                    ____________
for (int j = i+1; j < list.length; j++) {
  if (currentMin > list[j]) {
    currentMin = list[j];
                    ____________
  }
}
```

```
for (int i = 0; i < listSize; i++) {
  select the smallest element in list[i..listSize-1];
  swap the smallest with list[i], if necessary;
  // list[i] is in its correct position.
  // The next iteration apply on list[i..listSize-1]
}
```

Expand

```
double currentMin = list[i];
int currentMinIndex = i;
for (int j = i; j < list.length; j++) {
  if (currentMin > list[j]) {
    currentMin = list[j];
    currentMinIndex = j;
  }
}
```




```
for (int i = 0; i < listSize; i++) {
  select the smallest element in list[i..listSize-1];
  swap the smallest with list[i], if necessary;
  // list[i] is in its correct position.
  // The next iteration apply on list[i..listSize-1]
}
```

Expand

```
if (currentMinIndex != i) {
  list[currentMinIndex] = list[i];
  list[i] = currentMin;
}
```



Wrap it in a Method

```
/** The method for sorting the numbers */
public static void selectionSort(double[] list) {
  for (int i = 0; i < list.length; i++) {
    // Find the minimum in the list[i..list.length-1]
    double currentMin = list[i];
    int currentMinIndex = i;
    for (int j = i + 1; j < list.length; j++) {
      if (currentMin > list[j]) {
        currentMin = list[j];
        currentMinIndex = j;
      }
    }

    // Swap list[i] with list[currentMinIndex] if necessary;
    if (currentMinIndex != i) {
      list[currentMinIndex] = list[i];
      list[i] = currentMin;
    }
  }
}
```

Invoke it

selectionSort(yourList)



The **Arrays** Class

The **java.util.Arrays** class contains useful methods for common array operations such as sorting and searching.



The **Arrays** Class (cont.)

Arrays.toString(list) method can be used to return a string representation for the list.

```
int[] list = {2, 4, 7, 10};
System.out.println(Arrays.toString(list));
```

displays **[2, 4, 7, 10]**.

• Arrays.equals(list1, list2) checks whether two arrays are strictly equal

```
int[] list1 = {2, 4, 7, 10};
int[] list2 = {2, 4, 7, 10};
int[] list3 = {4, 2, 7, 10};
System.out.println(java.util.Arrays.equals(list1,
list2)); // true
System.out.println(java.util.Arrays.equals(list2,
list3)); // false
```



The Arrays. binarySearch Method

Since binary search is frequently used in programming, Java provides several overloaded binarySearch methods for searching a key in an array of int, double, char, short, long, and float in the java.util.Arrays class. For example, the following code searches the keys in an array of numbers and an array of characters.

```
int[] list = {2, 4, 7, 10, 11, 45, 50, 59, 60, 66, 69, 70, 79};
System.out.println("Index is " +
java.util.Arrays.binarySearch(list, 11));
```

Return is 4

```
char[] chars = {'a', 'c', 'g', 'x', 'y', 'z'};
System.out.println("Index is " +
java.util.Arrays.binarySearch(chars, 't'));
```

Return is –4 (insertion point is 3, so return is -3-1)

For the binarySearch method to work, the array must be pre-sorted in increasing order.



The Arrays.sort Method

Since sorting is frequently used in programming, Java provides several overloaded sort methods for sorting an array of int, double, char, short, long, and float in the java.util.Arrays class. For example, the following code sorts an array of numbers and an array of characters.

```
double[] numbers = {6.0, 4.4, 1.9, 2.9, 3.4, 3.5};
java.util.Arrays.sort(numbers);

char[] chars = {'a', 'A', '4', 'F', 'D', 'P'};
java.util.Arrays.sort(chars);
```

Java 8 now provides Arrays.parallelSort(list) that utilizes the multicore for fast sorting.

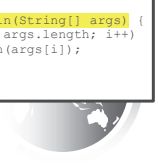


Command-Line Arguments

You can call a regular method by passing actual parameters. Can you pass arguments to main? Of course, yes. For example, the main method in class B is invoked by a method in A, as shown below:

```
public class A {
  public static void main(String[] args) {
    String[] strings = {"New York",
      "Boston", "Atlanta"};
    B.main(strings);
  }
}
```

```
class B {
  public static void main(String[] args) {
    for (int i = 0; i < args.length; i++)
      System.out.println(args[i]);
  }
}
```



Command-Line Parameters

```
class TestMain {
  public static void main(String[] args) {
  ...
  }
}
```

```
java TestMain arg0 arg1 arg2 ... argn
```



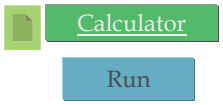
Processing Command-Line Parameters

In the main method, get the arguments from `args[0], args[1], ..., args[n]`, which corresponds to `arg0, arg1, ..., argn` in the command line.



Problem: Calculator

- Objective: Write a program that will perform binary operations on integers. The program receives three parameters: an operator and two integers.



java Calculator 2 + 3
java Calculator 2 - 3
java Calculator 2 / 3
java Calculator 2 . 3

